

Sample Spaces

A set S that consists of all possible outcomes of a random experiment is called a *sample space*, and each outcome is called a *sample point*. Often there will be more than one sample space that can describe outcomes of an experiment, but there is usually only one that will provide the most information.

EXAMPLE 1.6 If we toss a die, one sample space, or set of all possible outcomes, is given by $\{1, 2, 3, 4, 5, 6\}$ while another is $\{\text{odd}, \text{even}\}$. It is clear, however, that the latter would not be adequate to determine, for example, whether an outcome is divisible by 3.

It is often useful to portray a sample space graphically. In such cases it is desirable to use numbers in place of letters whenever possible.

EXAMPLE 1.7 If we toss a coin twice and use 0 to represent tails and 1 to represent heads, the sample space (see Example 1.3) can be portrayed by points as in Fig. 1-1 where, for example, $(0, 1)$ represents tails on first toss and heads on second toss, i.e., TH .

Random Experiments

We are all familiar with the importance of experiments in science and engineering. Experimentation is useful to us because we can assume that if we perform certain experiments under very nearly identical conditions, we will arrive at results that are essentially the same. In these circumstances, we are able to control the value of the variables that affect the outcome of the experiment.

However, in some experiments, we are not able to ascertain or control the value of certain variables so that the results will vary from one performance of the experiment to the next even though most of the conditions are the same. These experiments are described as *random*. The following are some examples.

EXAMPLE 1.1 If we toss a coin, the result of the experiment is that it will either come up “tails,” symbolized by T (or 0), or “heads,” symbolized by H (or 1), i.e., one of the elements of the set $\{H, T\}$ (or $\{0, 1\}$).

EXAMPLE 1.2 If we toss a die, the result of the experiment is that it will come up with one of the numbers in the set $\{1, 2, 3, 4, 5, 6\}$.

EXAMPLE 1.3 If we toss a coin twice, there are four results possible, as indicated by $\{HH, HT, TH, TT\}$, i.e., both heads, heads on first and tails on second, etc.

EXAMPLE 1.4 If we are making bolts with a machine, the result of the experiment is that some may be defective. Thus when a bolt is made, it will be a member of the set $\{\text{defective}, \text{nondefective}\}$.

EXAMPLE 1.5 If an experiment consists of measuring “lifetimes” of electric light bulbs produced by a company, then the result of the experiment is a time t in hours that lies in some interval—say, $0 \leq t \leq 4000$ —where we assume that no bulb lasts more than 4000 hours.

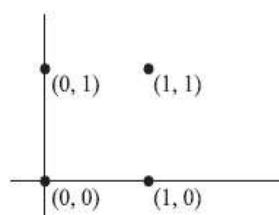


Fig. 1-1

If a sample space has a finite number of points, as in Example 1.7, it is called a *finite sample space*. If it has as many points as there are natural numbers $1, 2, 3, \dots$, it is called a *countably infinite sample space*. If it has as many points as there are in some interval on the x axis, such as $0 \leq x \leq 1$, it is called a *noncountably infinite sample space*. A sample space that is finite or countably infinite is often called a *discrete sample space*, while one that is noncountably infinite is called a *nondiscrete sample space*.

Events

An *event* is a subset A of the sample space S , i.e., it is a set of possible outcomes. If the outcome of an experiment is an element of A , we say that the event A *has occurred*. An event consisting of a single point of S is often called a *simple* or *elementary event*.

EXAMPLE 1.8 If we toss a coin twice, the event that only one head comes up is the subset of the sample space that consists of points $(0, 1)$ and $(1, 0)$, as indicated in Fig. 1-2.

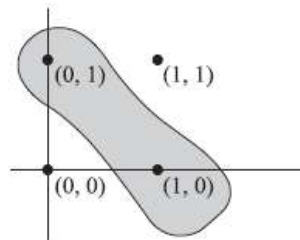


Fig. 1-2

As particular events, we have S itself, which is the *sure* or *certain event* since an element of S must occur, and the empty set $\emptyset$, which is called the *impossible event* because an element of $\emptyset$ cannot occur.

By using set operations on events in S , we can obtain other events in S . For example, if A and B are events, then

1. $A \cup B$ is the event “either A or B or both.” $A \cup B$ is called the *union* of A and B .
2. $A \cap B$ is the event “both A and B .” $A \cap B$ is called the *intersection* of A and B .
3. A' is the event “not A .” A' is called the *complement* of A .
4. $A - B = A \cap B'$ is the event “ A but not B .” In particular, $A' = S - A$.

If the sets corresponding to events A and B are disjoint, i.e., $A \cap B = \emptyset$, we often say that the events are *mutually exclusive*. This means that they cannot both occur. We say that a collection of events $A_1, A_2, \dots, A_n$ is *mutually exclusive* if every pair in the collection is mutually exclusive.

EXAMPLE 1.9 Referring to the experiment of tossing a coin twice, let A be the event “at least one head occurs” and B the event “the second toss results in a tail.” Then $A = \{HT, TH, HH\}$, $B = \{HT, TT\}$, and so we have

$$\begin{aligned} A \cup B &= \{HT, TH, HH, TT\} = S & A \cap B &= \{HT\} \\ A' &= \{TT\} & A - B &= \{TH, HH\} \end{aligned}$$

The Concept of Probability

In any random experiment there is always uncertainty as to whether a particular event will or will not occur. As a measure of the *chance*, or *probability*, with which we can expect the event to occur, it is convenient to assign a number between 0 and 1. If we are sure or certain that the event will occur, we say that its probability is 100% or 1, but if we are sure that the event will not occur, we say that its probability is zero. If, for example, the probability is $\frac{1}{4}$, we would say that there is a 25% chance it will occur and a 75% chance that it will not occur. Equivalently, we can say that the *odds* against its occurrence are 75% to 25%, or 3 to 1.

There are two important procedures by means of which we can estimate the probability of an event.

1. **CLASSICAL APPROACH.** If an event can occur in h different ways out of a total number of n possible ways, all of which are equally likely, then the probability of the event is h/n .

EXAMPLE 1.10 Suppose we want to know the probability that a head will turn up in a single toss of a coin. Since there are two equally likely ways in which the coin can come up—namely, heads and tails (assuming it does not roll away or stand on its edge)—and of these two ways a head can arise in only one way, we reason that the required probability is $1/2$. In arriving at this, we assume that the coin is *fair*, i.e., not *loaded* in any way.

2. **FREQUENCY APPROACH.** If after n repetitions of an experiment, where n is very large, an event is observed to occur in h of these, then the probability of the event is h/n . This is also called the *empirical probability* of the event.

EXAMPLE 1.11 If we toss a coin 1000 times and find that it comes up heads 532 times, we estimate the probability of a head coming up to be $532/1000 = 0.532$.

Both the classical and frequency approaches have serious drawbacks, the first because the words “equally likely” are vague and the second because the “large number” involved is vague. Because of these difficulties, mathematicians have been led to an *axiomatic approach* to probability.

The Axioms of Probability

Suppose we have a sample space S . If S is discrete, all subsets correspond to events and conversely, but if S is nondiscrete, only special subsets (called *measurable*) correspond to events. To each event A in the class C of events, we associate a real number $P(A)$. Then P is called a *probability function*, and $P(A)$ the *probability* of the event A , if the following axioms are satisfied.

Axiom 1 For every event A in the class C ,

$$P(A) \geq 0 \quad (1)$$

Axiom 2 For the sure or certain event S in the class C ,

$$P(S) = 1 \quad (2)$$

Axiom 3 For any number of mutually exclusive events $A_1, A_2, \dots$, in the class C ,

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots \quad (3)$$

In particular, for two mutually exclusive events A_1, A_2 ,

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) \quad (4)$$

Some Important Theorems on Probability

From the above axioms we can now prove various theorems on probability that are important in further work.

Theorem 1-1 If $A_1 \subset A_2$, then $P(A_1) \leq P(A_2)$ and $P(A_2 - A_1) = P(A_2) - P(A_1)$.

Theorem 1-2 For every event A ,

$$0 \leq P(A) \leq 1, \quad (5)$$

i.e., a probability is between 0 and 1.

Theorem 1-3 $P(\emptyset) = 0$ (6)

i.e., the impossible event has probability zero.

Theorem 1-4 If A' is the complement of A , then

$$P(A') = 1 - P(A) \quad (7)$$

Theorem 1-5 If $A = A_1 \cup A_2 \cup \dots \cup A_n$, where $A_1, A_2, \dots, A_n$ are mutually exclusive events, then

$$P(A) = P(A_1) + P(A_2) + \dots + P(A_n) \quad (8)$$

In particular, if $A = S$, the sample space, then

$$P(A_1) + P(A_2) + \dots + P(A_n) = 1 \quad (9)$$

Theorem 1-6 If A and B are any two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (10)$$

More generally, if A_1, A_2, A_3 are any three events, then

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) &= P(A_1) + P(A_2) + P(A_3) \\ &\quad - P(A_1 \cap A_2) - P(A_2 \cap A_3) - P(A_3 \cap A_1) \\ &\quad + P(A_1 \cap A_2 \cap A_3) \end{aligned} \quad (11)$$

Generalizations to n events can also be made.

Theorem 1-7 For any events A and B ,

$$P(A) = P(A \cap B) + P(A \cap B') \quad (12)$$

Theorem 1-8 If an event A must result in the occurrence of one of the mutually exclusive events $A_1, A_2, \dots, A_n$, then

$$P(A) = P(A \cap A_1) + P(A \cap A_2) + \dots + P(A \cap A_n) \quad (13)$$

Assignment of Probabilities

If a sample space S consists of a finite number of outcomes $a_1, a_2, \dots, a_n$, then by Theorem 1-5,

$$P(A_1) + P(A_2) + \dots + P(A_n) = 1 \quad (14)$$

where $A_1, A_2, \dots, A_n$ are elementary events given by $A_i = \{a_i\}$.

It follows that we can arbitrarily choose any nonnegative numbers for the probabilities of these simple events as long as (14) is satisfied. In particular, if we assume *equal probabilities* for all simple events, then

$$P(A_k) = \frac{1}{n}, \quad k = 1, 2, \dots, n \quad (15)$$

and if A is any event made up of h such simple events, we have

$$P(A) = \frac{h}{n} \quad (16)$$

This is equivalent to the classical approach to probability given on page 5. We could of course use other procedures for assigning probabilities, such as the frequency approach of page 5.

Assigning probabilities provides a *mathematical model*, the success of which must be tested by experiment in much the same manner that theories in physics or other sciences must be tested by experiment.

EXAMPLE 1.12 A single die is tossed once. Find the probability of a 2 or 5 turning up.

The sample space is $S = \{1, 2, 3, 4, 5, 6\}$. If we assign equal probabilities to the sample points, i.e., if we assume that the die is fair, then

$$P(1) = P(2) = \dots = P(6) = \frac{1}{6}$$

The event that either 2 or 5 turns up is indicated by $2 \cup 5$. Therefore,

$$P(2 \cup 5) = P(2) + P(5) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

Conditional Probability

Let A and B be two events (Fig. 1-3) such that $P(A) > 0$. Denote by $P(B|A)$ the probability of B given that A has occurred. Since A is known to have occurred, it becomes the new sample space replacing the original S . From this we are led to the definition

$$P(B|A) \equiv \frac{P(A \cap B)}{P(A)} \quad (17)$$

or

$$P(A \cap B) \equiv P(A) P(B|A) \quad (18)$$

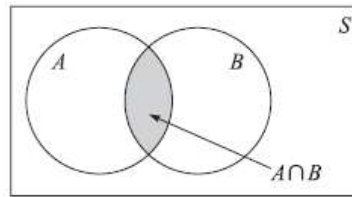


Fig. 1-3

In words, (18) says that the probability that both A and B occur is equal to the probability that A occurs times the probability that B occurs given that A has occurred. We call $P(B|A)$ the *conditional probability* of B given A , i.e., the probability that B will occur given that A has occurred. It is easy to show that conditional probability satisfies the axioms on page 5.

EXAMPLE 1.13 Find the probability that a single toss of a die will result in a number less than 4 if (a) no other information is given and (b) it is given that the toss resulted in an odd number.

(a) Let B denote the event {less than 4}. Since B is the union of the events 1, 2, or 3 turning up, we see by Theorem 1-5 that

$$P(B) = P(1) + P(2) + P(3) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

assuming equal probabilities for the sample points.

(b) Letting A be the event {odd number}, we see that $P(A) = \frac{3}{6} = \frac{1}{2}$. Also $P(A \cap B) = \frac{2}{6} = \frac{1}{3}$. Then

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1/3}{1/2} = \frac{2}{3}$$

Hence, the added knowledge that the toss results in an odd number raises the probability from $1/2$ to $2/3$.

Theorems on Conditional Probability

Theorem 1-9 For any three events A_1, A_2, A_3 , we have

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) P(A_2|A_1) P(A_3|A_1 \cap A_2) \quad (19)$$

In words, the probability that A_1 and A_2 and A_3 all occur is equal to the probability that A_1 occurs times the probability that A_2 occurs given that A_1 has occurred times the probability that A_3 occurs given that both A_1 and A_2 have occurred. The result is easily generalized to n events.

Theorem 1-10 If an event A must result in one of the mutually exclusive events $A_1, A_2, \dots, A_n$, then

$$P(A) = P(A_1) P(A|A_1) + P(A_2) P(A|A_2) + \dots + P(A_n) P(A|A_n) \quad (20)$$

Independent Events

If $P(B|A) = P(B)$, i.e., the probability of B occurring is not affected by the occurrence or non-occurrence of A , then we say that A and B are *independent events*. This is equivalent to

$$P(A \cap B) = P(A)P(B) \quad (21)$$

as seen from (18). Conversely, if (21) holds, then A and B are independent.

We say that three events A_1, A_2, A_3 are *independent* if they are pairwise independent:

$$P(A_j \cap A_k) = P(A_j)P(A_k) \quad j \neq k \quad \text{where } j, k = 1, 2, 3 \quad (22)$$

and

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3) \quad (23)$$

Note that neither (22) nor (23) is by itself sufficient. Independence of more than three events is easily defined.

Bayes' Theorem or Rule

Suppose that $A_1, A_2, \dots, A_n$ are mutually exclusive events whose union is the sample space S , i.e., one of the events must occur. Then if A is any event, we have the following important theorem:

Theorem 1-II (Bayes' Rule):

$$P(A_k|A) = \frac{P(A_k)P(A|A_k)}{\sum_{j=1}^n P(A_j)P(A|A_j)} \quad (24)$$

This enables us to find the probabilities of the various events $A_1, A_2, \dots, A_n$ that can *cause* A to occur. For this reason Bayes' theorem is often referred to as a *theorem on the probability of causes*.

Random Variables

Suppose that to each point of a sample space we assign a number. We then have a *function* defined on the sample space. This function is called a *random variable* (or *stochastic variable*) or more precisely a *random function* (*stochastic function*). It is usually denoted by a capital letter such as X or Y . In general, a random variable has some specified physical, geometrical, or other significance.

EXAMPLE 2.1 Suppose that a coin is tossed twice so that the sample space is $S = \{HH, HT, TH, TT\}$. Let X represent the number of heads that can come up. With each sample point we can associate a number for X as shown in Table 2-1. Thus, for example, in the case of HH (i.e., 2 heads), $X = 2$ while for TH (1 head), $X = 1$. It follows that X is a random variable.

Table 2-1

Sample Point	HH	HT	TH	TT
X	2	1	1	0

It should be noted that many other random variables could also be defined on this sample space, for example, the square of the number of heads or the number of heads minus the number of tails.

A random variable that takes on a finite or countably infinite number of values (see page 4) is called a *discrete random variable* while one which takes on a noncountably infinite number of values is called a *nondiscrete random variable*.

Discrete Probability Distributions

Let X be a discrete random variable, and suppose that the possible values that it can assume are given by $x_1, x_2, x_3, \dots$, arranged in some order. Suppose also that these values are assumed with probabilities given by

$$P(X = x_k) = f(x_k) \quad k = 1, 2, \dots \quad (1)$$

It is convenient to introduce the *probability function*, also referred to as *probability distribution*, given by

$$P(X = x) = f(x) \quad (2)$$

For $x = x_k$, this reduces to (1) while for other values of x , $f(x) = 0$.

In general, $f(x)$ is a probability function if

1. $f(x) \geq 0$
2. $\sum_x f(x) = 1$

where the sum in 2 is taken over all possible values of x .

EXAMPLE 2.2 Find the probability function corresponding to the random variable X of Example 2.1. Assuming that the coin is fair, we have

$$P(HH) = \frac{1}{4} \quad P(HT) = \frac{1}{4} \quad P(TH) = \frac{1}{4} \quad P(TT) = \frac{1}{4}$$

Then

$$P(X = 0) = P(TT) = \frac{1}{4}$$

$$P(X = 1) = P(HT \cup TH) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$P(X = 2) = P(HH) = \frac{1}{4}$$

The probability function is thus given by Table 2-2.

Table 2-2

x	0	1	2
$f(x)$	1/4	1/2	1/4

Distribution Functions for Random Variables

The *cumulative distribution function*, or briefly the *distribution function*, for a random variable X is defined by

$$F(x) = P(X \leq x) \quad (3)$$

where x is any real number, i.e., $-\infty < x < \infty$.

The distribution function $F(x)$ has the following properties:

1. $F(x)$ is nondecreasing [i.e., $F(x) \leq F(y)$ if $x \leq y$].
2. $\lim_{x \rightarrow -\infty} F(x) = 0$; $\lim_{x \rightarrow \infty} F(x) = 1$.
3. $F(x)$ is continuous from the right [i.e., $\lim_{h \rightarrow 0^+} F(x + h) = F(x)$ for all x].

Distribution Functions for Discrete Random Variables

The distribution function for a discrete random variable X can be obtained from its probability function by noting that, for all x in $(-\infty, \infty)$,

$$F(x) = P(X \leq x) = \sum_{u \leq x} f(u) \quad (4)$$

where the sum is taken over all values u taken on by X for which $u \leq x$.

If X takes on only a finite number of values $x_1, x_2, \dots, x_n$, then the distribution function is given by

$$F(x) = \begin{cases} 0 & -\infty < x < x_1 \\ f(x_1) & x_1 \leq x < x_2 \\ f(x_1) + f(x_2) & x_2 \leq x < x_3 \\ \vdots & \vdots \\ f(x_1) + \dots + f(x_n) & x_n \leq x < \infty \end{cases} \quad (5)$$

EXAMPLE 2.3 (a) Find the distribution function for the random variable X of Example 2.2. (b) Obtain its graph.

(a) The distribution function is

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ \frac{1}{4} & 0 \leq x < 1 \\ \frac{3}{4} & 1 \leq x < 2 \\ 1 & 2 \leq x < \infty \end{cases}$$

(b) The graph of $F(x)$ is shown in Fig. 2-1.

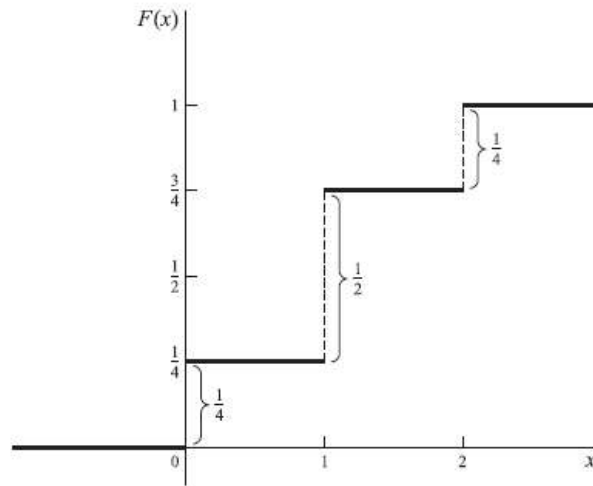


Fig. 2-1

The following things about the above distribution function, which are true in general, should be noted.

1. The magnitudes of the jumps at 0, 1, 2 are $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ which are precisely the probabilities in Table 2-2. This fact enables one to obtain the probability function from the distribution function.
2. Because of the appearance of the graph of Fig. 2-1, it is often called a *staircase function* or *step function*. The value of the function at an integer is obtained from the higher step; thus the value at 1 is $\frac{3}{4}$ and not $\frac{1}{4}$. This is expressed mathematically by stating that the distribution function is *continuous from the right* at 0, 1, 2.
3. As we proceed from left to right (i.e. going *upstairs*), the distribution function either remains the same or increases, taking on values from 0 to 1. Because of this, it is said to be a *monotonically increasing function*.

It is clear from the above remarks and the properties of distribution functions that the probability function of a discrete random variable can be obtained from the distribution function by noting that

$$f(x) = F(x) - \lim_{u \rightarrow x^-} F(u). \quad (6)$$

Continuous Random Variables

A nondiscrete random variable X is said to be *absolutely continuous*, or simply *continuous*, if its distribution function may be represented as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du \quad (-\infty < x < \infty) \quad (7)$$

where the function $f(x)$ has the properties

1. $f(x) \geq 0$
2. $\int_{-\infty}^{\infty} f(x) dx = 1$

It follows from the above that if X is a continuous random variable, then the probability that X takes on any one particular value is zero, whereas the *interval probability* that X lies *between two different values*, say, a and b , is given by

$$P(a < X < b) = \int_a^b f(x) dx \quad (8)$$

EXAMPLE 2.4 If an individual is selected at random from a large group of adult males, the probability that his height X is precisely 68 inches (i.e., 68.000 ... inches) would be zero. However, there is a probability greater than zero that X is between 67.000 ... inches and 68.500 ... inches, for example.

A function $f(x)$ that satisfies the above requirements is called a *probability function* or *probability distribution* for a continuous random variable, but it is more often called a *probability density function* or simply *density function*. Any function $f(x)$ satisfying Properties 1 and 2 above will automatically be a density function, and required probabilities can then be obtained from (8).

EXAMPLE 2.5 (a) Find the constant c such that the function

$$f(x) = \begin{cases} cx^2 & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

is a density function, and (b) compute $P(1 < X < 2)$.

(a) Since $f(x)$ satisfies Property 1 if $c \geq 0$, it must satisfy Property 2 in order to be a density function. Now

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^3 cx^2 dx = \left. \frac{cx^3}{3} \right|_0^3 = 9c$$

and since this must equal 1, we have $c = 1/9$.

(b)
$$P(1 < X < 2) = \int_1^2 \frac{1}{9} x^2 dx = \left. \frac{x^3}{27} \right|_1^2 = \frac{8}{27} - \frac{1}{27} = \frac{7}{27}$$

In case $f(x)$ is continuous, which we shall assume unless otherwise stated, the probability that X is equal to any particular value is zero. In such case we can replace either or both of the signs $<$ in (8) by $\leq$. Thus, in Example 2.5,

$$P(1 \leq X \leq 2) = P(1 \leq X < 2) = P(1 < X \leq 2) = P(1 < X < 2) = \frac{7}{27}$$

EXAMPLE 2.6 (a) Find the distribution function for the random variable of Example 2.5. (b) Use the result of (a) to find $P(1 < x \leq 2)$.

(a) We have

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$$

If $x < 0$, then $F(x) = 0$. If $0 \leq x < 3$, then

$$F(x) = \int_0^x f(u) du = \int_0^x \frac{1}{9} u^2 du = \frac{x^3}{27}$$

If $x \geq 3$, then

$$F(x) = \int_0^3 f(u) du + \int_3^x f(u) du = \int_0^3 \frac{1}{9} u^2 du + \int_3^x 0 du = 1$$

Thus the required distribution function is

$$F(x) = \begin{cases} 0 & x < 0 \\ x^3/27 & 0 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

Note that $F(x)$ increases monotonically from 0 to 1 as is required for a distribution function. It should also be noted that $F(x)$ in this case is continuous.



(b) We have

$$\begin{aligned} P(1 < X \leq 2) &= P(X \leq 2) - P(X \leq 1) \\ &= F(2) - F(1) \\ &= \frac{2^3}{27} - \frac{1^3}{27} = \frac{7}{27} \end{aligned}$$

as in Example 2.5.

The probability that X is between x and $x + \Delta x$ is given by

$$P(x \leq X \leq x + \Delta x) = \int_x^{x+\Delta x} f(u) du \quad (9)$$

so that if Δx is small, we have approximately

$$P(x \leq X \leq x + \Delta x) = f(x)\Delta x \quad (10)$$

We also see from (7) on differentiating both sides that

$$\frac{dF(x)}{dx} = f(x) \quad (11)$$

at all points where $f(x)$ is continuous; i.e., the derivative of the distribution function is the density function.

It should be pointed out that random variables exist that are neither discrete nor continuous. It can be shown that the random variable X with the following distribution function is an example.

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{x}{2} & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

In order to obtain (11), we used the basic property

$$\frac{d}{dx} \int_a^x f(u) du = f(x) \quad (12)$$

which is one version of the Fundamental Theorem of Calculus.

Graphical Interpretations

If $f(x)$ is the density function for a random variable X , then we can represent $y = f(x)$ graphically by a curve as in Fig. 2-2. Since $f(x) \geq 0$, the curve cannot fall below the x axis. The entire area bounded by the curve and the x axis must be 1 because of Property 2 on page 36. Geometrically the probability that X is between a and b , i.e., $P(a < X < b)$, is then represented by the area shown shaded, in Fig. 2-2.

The distribution function $F(x) = P(X \leq x)$ is a monotonically increasing function which increases from 0 to 1 and is represented by a curve as in Fig. 2-3.

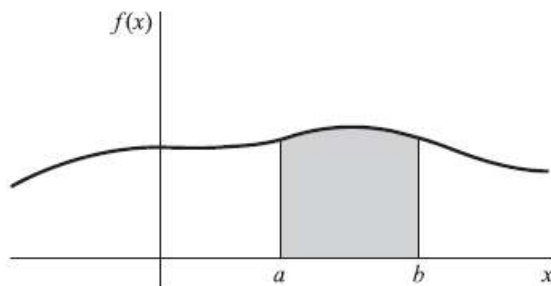


Fig. 2-2

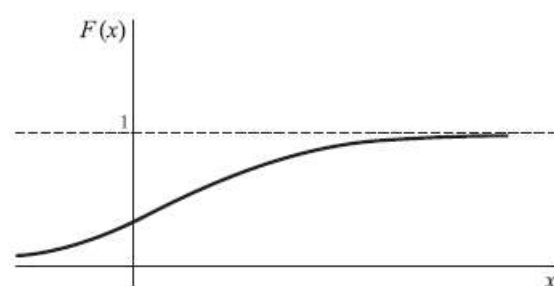


Fig. 2-3

Joint Distributions

The above ideas are easily generalized to two or more random variables. We consider the typical case of two random variables that are either both discrete or both continuous. In cases where one variable is discrete and the other continuous, appropriate modifications are easily made. Generalizations to more than two variables can also be made.

1. DISCRETE CASE. If X and Y are two discrete random variables, we define the *joint probability function* of X and Y by

$$P(X = x, Y = y) = f(x, y) \quad (13)$$

where 1. $f(x, y) \geq 0$

$$2. \sum_x \sum_y f(x, y) = 1$$

i.e., the sum over all values of x and y is 1.

Suppose that X can assume any one of m values $x_1, x_2, \dots, x_m$ and Y can assume any one of n values $y_1, y_2, \dots, y_n$. Then the probability of the event that $X = x_j$ and $Y = y_k$ is given by

$$P(X = x_j, Y = y_k) = f(x_j, y_k) \quad (14)$$

A joint probability function for X and Y can be represented by a *joint probability table* as in Table 2-3. The probability that $X = x_j$ is obtained by adding all entries in the row corresponding to x_i and is given by

$$P(X = x_j) = f_1(x_j) = \sum_{k=1}^n f(x_j, y_k) \quad (15)$$

Table 2-3

$\begin{matrix} Y \\ X \end{matrix}$	y_1	y_2	$\dots$	y_n	Totals ↓
x_1	$f(x_1, y_1)$	$f(x_1, y_2)$	$\dots$	$f(x_1, y_n)$	$f_1(x_1)$
x_2	$f(x_2, y_1)$	$f(x_2, y_2)$	$\dots$	$f(x_2, y_n)$	$f_1(x_2)$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
x_m	$f(x_m, y_1)$	$f(x_m, y_2)$	$\dots$	$f(x_m, y_n)$	$f_1(x_m)$
Totals →	$f_2(y_1)$	$f_2(y_2)$	$\dots$	$f_2(y_n)$	1 ← Grand Total

For $j = 1, 2, \dots, m$, these are indicated by the entry totals in the extreme right-hand column or margin of Table 2-3. Similarly the probability that $Y = y_k$ is obtained by adding all entries in the column corresponding to y_k and is given by

$$P(Y = y_k) = f_2(y_k) = \sum_{j=1}^m f(x_j, y_k) \quad (16)$$

For $k = 1, 2, \dots, n$, these are indicated by the entry totals in the bottom row or margin of Table 2-3.

Because the probabilities (15) and (16) are obtained from the margins of the table, we often refer to $f_1(x_j)$ and $f_2(y_k)$ [or simply $f_1(x)$ and $f_2(y)$] as the *marginal probability functions* of X and Y , respectively.

It should also be noted that

$$\sum_{j=1}^m f_1(x_j) = 1 \quad \sum_{k=1}^n f_2(y_k) = 1 \quad (17)$$

which can be written

$$\sum_{j=1}^m \sum_{k=1}^n f(x_j, y_k) = 1 \quad (18)$$

This is simply the statement that the total probability of all entries is 1. The *grand total* of 1 is indicated in the lower right-hand corner of the table.

The *joint distribution function* of X and Y is defined by

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{u \leq x} \sum_{v \leq y} f(u, v) \quad (19)$$

In Table 2-3, $F(x, y)$ is the sum of all entries for which $x_j \leq x$ and $y_k \leq y$.

2. CONTINUOUS CASE. The case where both variables are continuous is obtained easily by analogy with the discrete case on replacing sums by integrals. Thus the *joint probability function* for the random variables X and Y (or, as it is more commonly called, the *joint density function* of X and Y) is defined by

1. $f(x, y) \geq 0$
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

Graphically $z = f(x, y)$ represents a surface, called the *probability surface*, as indicated in Fig. 2-4. The total volume bounded by this surface and the xy plane is equal to 1 in accordance with Property 2 above. The probability that X lies between a and b while Y lies between c and d is given graphically by the shaded volume of Fig. 2-4 and mathematically by

$$P(a < X < b, c < Y < d) = \int_{x=a}^b \int_{y=c}^d f(x, y) dx dy \quad (20)$$

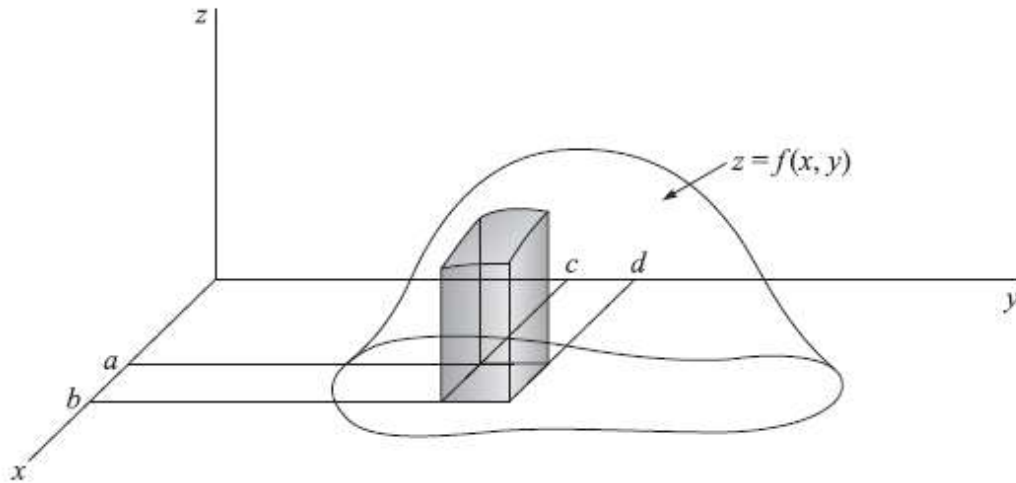


Fig. 2-4

More generally, if A represents any event, there will be a region $\mathcal{R}_A$ of the xy plane that corresponds to it. In such case we can find the probability of A by performing the integration over $\mathcal{R}_A$, i.e.,

$$P(A) = \iint_{\mathcal{R}_A} f(x, y) dx dy \quad (21)$$

The *joint distribution function* of X and Y in this case is defined by

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{u=-\infty}^x \int_{v=-\infty}^y f(u, v) du dv \quad (22)$$

It follows in analogy with (11), page 38, that

$$\frac{\partial^2 F}{\partial x \partial y} = f(x, y) \quad (23)$$

i.e., the density function is obtained by differentiating the distribution function with respect to x and y .

From (22) we obtain

$$P(X \leq x) = F_1(x) = \int_{u=-\infty}^x \int_{v=-\infty}^{\infty} f(u, v) du dv \quad (24)$$

$$P(Y \leq y) = F_2(y) = \int_{u=-\infty}^{\infty} \int_{v=-\infty}^y f(u, v) du dv \quad (25)$$

We call (24) and (25) the *marginal distribution functions*, or simply the *distribution functions*, of X and Y , respectively. The derivatives of (24) and (25) with respect to x and y are then called the *marginal density functions*, or simply the *density functions*, of X and Y and are given by

$$f_1(x) = \int_{v=-\infty}^{\infty} f(x, v) dv \quad f_2(y) = \int_{u=-\infty}^{\infty} f(u, y) du \quad (26)$$

Numerical Analysis: Solved Examples

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April 25, 2025

1 Error Analysis

Example 1: Absolute and Relative Errors

Problem: Given the true value $T_{\text{true}} = 3.1415926535$ and the approximate value $T_{\text{approx}} = 3.1428571429$, compute the absolute and relative errors.

Solution:

- Absolute Error:

$$E_{\text{abs}} = |T_{\text{true}} - T_{\text{approx}}| = |3.1415926535 - 3.1428571429| = 0.0012644894$$

- Relative Error:

$$E_{\text{rel}} = \frac{E_{\text{abs}}}{|T_{\text{true}}|} = \frac{0.0012644894}{3.1415926535} \approx 0.0004022 \quad \text{or } 0.04022\%$$

Example 2: Error Propagation

Problem: A radius $r = 5$ cm has an error $\Delta r = \pm 0.1$ cm. Estimate the maximum error in the area $A = \pi r^2$.

Solution:

- Use differentials: $\Delta A \approx \left| \frac{dA}{dr} \right| \Delta r = |2\pi r| \Delta r$
- Substituting values: $\Delta A = 2\pi(5)(0.1) = \pi \text{ cm}^2 \approx 3.1416 \text{ cm}^2$
- Relative error in area:

$$\frac{\Delta A}{A} = \frac{2\Delta r}{r} = \frac{2(0.1)}{5} = 0.04 \quad \text{or } 4\%$$

2 Newton-Raphson Method

Example 3: Finding the Root of $f(x) = x^3 - x - 1$

Problem: Use the Newton-Raphson method starting from $x_0 = 1$ to find a root of $f(x) = x^3 - x - 1$.

Solution:

- $f(x) = x^3 - x - 1, \quad f'(x) = 3x^2 - 1$

- Iteration 1:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \frac{1^3 - 1 - 1}{3(1)^2 - 1} = 1 - \frac{-1}{2} = 1.5$$

- Iteration 2:

$$f(1.5) = 1.5^3 - 1.5 - 1 = 0.875, \quad f'(1.5) = 3(1.5)^2 - 1 = 5.75$$

$$x_2 = 1.5 - \frac{0.875}{5.75} \approx 1.34783$$

- Iteration 3:

$$f(1.34783) \approx 0.100, \quad f'(1.34783) \approx 4.449$$

$$x_3 = 1.34783 - \frac{0.100}{4.449} \approx 1.32561$$

- Final result after 4 iterations: $x \approx 1.3247$

Example 4: Finding the Root of $f(x) = e^x - 2x - 2$

Problem: Use the Newton-Raphson method with $x_0 = 1$.

Solution:

- $f(x) = e^x - 2x - 2, \quad f'(x) = e^x - 2$

- Iteration 1:

$$f(1) = e^1 - 2 - 2 \approx 2.718 - 4 = -1.2817, \quad f'(1) = e^1 - 2 \approx 0.7183$$

$$x_1 = 1 - \frac{-1.2817}{0.7183} \approx 2.784$$

- Iteration 2:

$$f(2.784) \approx 8.582, \quad f'(2.784) \approx 14.15$$

$$x_2 = 2.784 - \frac{8.582}{14.15} \approx 2.178$$

- Iteration 3:

$$f(2.178) \approx 2.464, \quad f'(2.178) \approx 6.82$$

$$x_3 = 2.178 - \frac{2.464}{6.82} \approx 1.817$$

- Final result after 6 iterations: $x \approx 1.6783$

3 Lagrange Interpolation Polynomial

Example 5: Linear Interpolation

Problem: Interpolate the function through the points $(1, 2)$ and $(3, 5)$.

Solution:

- Basis polynomials:

$$L_0(x) = \frac{x-3}{1-3}, \quad L_1(x) = \frac{x-1}{3-1}$$

- Interpolating polynomial:

$$P_1(x) = 2L_0(x) + 5L_1(x) = 2 \cdot \frac{x-3}{-2} + 5 \cdot \frac{x-1}{2}$$

- Simplified:

$$P_1(x) = \frac{-2(x-3) + 5(x-1)}{2} = \frac{3x+1}{2}$$

Example 6: Quadratic Interpolation

Problem: Interpolate the function through $(0, 1)$, $(1, 2)$, and $(2, 5)$.

Solution:

- Basis polynomials:

$$L_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2}$$

$$L_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)} = -x(x-2)$$

$$L_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}$$

- Interpolating polynomial:

$$\begin{aligned} P_2(x) &= 1 \cdot L_0(x) + 2 \cdot L_1(x) + 5 \cdot L_2(x) \\ &= \frac{(x-1)(x-2)}{2} - 2x(x-2) + \frac{5x(x-1)}{2} \end{aligned}$$

- Simplified:

$$P_2(x) = x^2 + 1$$

Example 7: Cubic Interpolation

Problem: Interpolate the points $(-1, 1)$, $(0, 0)$, $(1, 1)$, and $(2, 4)$.

Solution: Constructing full basis polynomials $L_0(x), L_1(x), L_2(x), L_3(x)$ and applying them:

$$P_3(x) = 1 \cdot L_0(x) + 0 \cdot L_1(x) + 1 \cdot L_2(x) + 4 \cdot L_3(x)$$

(Simplified version yields: $P_3(x) = x^2$)

4 Newton Interpolation Polynomial

Formulas for Divided Differences

1. First Divided Difference (linear term):

$$f[x_i, x_j] = \frac{f(x_j) - f(x_i)}{x_j - x_i}$$

2. Second Divided Difference (quadratic term):

$$f[x_i, x_j, x_k] = \frac{f[x_j, x_k] - f[x_i, x_j]}{x_k - x_i}$$

3. n-th Divided Difference:

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}$$

Divided Differences Table

Given nodes $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, the divided differences table is structured as:

x	$f[\]$	$f[\ , \]$	$f[\ , \ , \]$	$\dots$
x_0	$f[x_0]$	$f[x_0, x_1]$	$f[x_0, x_1, x_2]$	$\dots$
x_1	$f[x_1]$	$f[x_1, x_2]$	$\dots$	
x_2	$f[x_2]$	$\dots$		
$\vdots$	$\vdots$			
x_n	$f[x_n]$			$f[x_0, \dots, x_n]$

- Column 1: Original function values $f[x_i] = y_i$
- Column 2: First divided differences $f[x_i, x_j]$
- Column 3: Second divided differences $f[x_i, x_j, x_k]$, and so on

Examples

Example: Linear Newton Polynomial

Problem: Construct the Newton polynomial for $(1, 3)$ and $(4, 7)$.

Solution:

1. Divided Differences Table:

x	$f[\]$	$f[\ , \]$
1	3	$\frac{7-3}{4-1} = \frac{4}{3}$
4	7	

2. Newton Polynomial:

$$P_1(x) = f[x_0] + f[x_0, x_1](x - x_0) = 3 + \frac{4}{3}(x - 1)$$

3. Simplified:

$$P_1(x) = \frac{4}{3}x + \frac{5}{3}$$

Example: Quadratic Newton Polynomial

Problem: Construct the Newton polynomial for $(2, 5)$, $(3, 7)$, and $(5, 11)$.

Solution:

1. Divided Differences Table:

x	$f[]$	$f[,]$	$f[, ,]$
2	5	$\frac{7-5}{3-2} = 2$	$\frac{2-2}{5-2} = 0$
3	7	$\frac{11-7}{5-3} = 2$	
5	11		

2. Newton Polynomial:

$$P_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

Substituting values:

$$P_2(x) = 5 + 2(x - 2) + 0(x - 2)(x - 3)$$

3. Simplified:

$$P_2(x) = 2x + 1 \quad (\text{Linear due to collinearity})$$

Example 8: Linear Newton Polynomial

Problem: Interpolate (1, 3) and (4, 7) using Newton's method.

Solution:

- Divided difference:

$$f[1, 4] = \frac{7 - 3}{4 - 1} = \frac{4}{3}$$

- Polynomial:

$$P_1(x) = 3 + \frac{4}{3}(x - 1)$$

Example 9: Quadratic Newton Polynomial

Problem: Interpolate (2, 5), (3, 7), and (5, 11).

Solution:

- First-order differences:

$$f[2, 3] = \frac{7 - 5}{3 - 2} = 2, \quad f[3, 5] = \frac{11 - 7}{5 - 3} = 2$$

- Second-order difference:

$$f[2, 3, 5] = \frac{2 - 2}{5 - 2} = 0$$

- Polynomial:

$$P_2(x) = 5 + 2(x - 2) + 0(x - 2)(x - 3) = 2x + 1$$

Example 10: Cubic Newton Polynomial

Problem: Construct the Newton interpolating polynomial for the points:

$$(-2, -8), \quad (0, 0), \quad (1, 1), \quad (3, 27)$$

1. Divided Differences Table

We denote:

$$f[x_0] = f(-2) = -8, \quad f[x_1] = f(0) = 0, \quad f[x_2] = f(1) = 1, \quad f[x_3] = f(3) = 27$$

First-order divided differences:

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{0 - (-8)}{0 - (-2)} = \frac{8}{2} = 4$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{1 - 0}{1 - 0} = 1$$

$$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2} = \frac{27 - 1}{3 - 1} = \frac{26}{2} = 13$$

Second-order divided differences:

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{1 - 4}{1 - (-2)} = \frac{-3}{3} = -1$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{13 - 1}{3 - 0} = \frac{12}{3} = 4$$

Third-order divided difference:

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{4 - (-1)}{3 - (-2)} = \frac{5}{5} = 1$$

Complete Divided Differences Table:

x	$f[x]$	$f[\ , \]$	$f[\ , \ , \]$	$f[\ , \ , \ , \]$
-2	-8	4	-1	1
0	0	1	4	
1	1	13		
3	27			

2. Newton Interpolating Polynomial

General form:

$$P_3(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2)$$

Substitute values:

$$P_3(x) = -8 + 4(x + 2) - 1(x + 2)(x - 0) + 1(x + 2)(x)(x - 1)$$

3. Simplify Step-by-Step

First term:

$$-8$$

Second term:

$$4(x + 2) = 4x + 8$$

Third term:

$$-1(x + 2)(x) = -x(x + 2) = -x^2 - 2x$$

Fourth term:

$$(x+2)(x)(x-1) = x(x+2)(x-1) = x[(x^2-x)+2(x-1)] = x(x^2-x+2x-2) = x(x^2+x-2) = x^3+x^2-2x$$

Now sum all parts:

$$P_3(x) = -8 + 4x + 8 - x^2 - 2x + x^3 + x^2 - 2x$$

Combine like terms:

$$P_3(x) = x^3 + (4x - 2x - 2x) + (-x^2 + x^2) + (-8 + 8) = x^3 + 0x^2 + 0x + 0 = x^3$$

Final Result:

$$\boxed{P_3(x) = x^3}$$

Conclusion: The interpolating polynomial that passes through the points $(-2, -8)$, $(0, 0)$, $(1, 1)$, and $(3, 27)$ is the monomial x^3 . This confirms the data lies exactly on the cubic curve $y = x^3$.

5 Homework Problems

1. Given $e \approx 2.718$, find the absolute and relative errors compared to $e_{\text{true}} = 2.7182818284$.
2. A cube has a side length 10 cm with error ± 0.05 cm. Estimate the maximum error in volume.
3. Use Newton-Raphson to find a root of $f(x) = x^3 - 3$, starting at $x_0 = 1$.
4. Use Newton-Raphson on $f(x) = \cos x - x$, starting at $x_0 = 0.5$.
5. Construct a linear Lagrange polynomial for points $(2, 4)$ and $(5, 7)$.
6. Find the quadratic Lagrange polynomial for $(-1, 2)$, $(0, 1)$, and $(1, 2)$.
7. Construct a cubic Lagrange polynomial for $(0, 0)$, $(1, 1)$, $(2, 8)$, and $(3, 27)$.
8. Use Newton's divided differences for a linear polynomial with $(3, 9)$ and $(6, 18)$.
9. Build a quadratic Newton polynomial for $(1, 1)$, $(2, 4)$, and $(4, 16)$.
10. Construct a cubic Newton polynomial for $(-2, -8)$, $(0, 0)$, $(1, 1)$, and $(3, 27)$.

References

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Numerical Analysis Lectures: An Introduction, Error Analysis, Newton-Raphson Method, and Interpolation

Based on Schaum's Outline of Numerical Analysis,
Second Edition by Francis Scheid (1989)

April 8, 2025

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1 Lecture 1: Introduction to Numerical Analysis and Error Analysis

1.1 Part 1.1: Introduction to Numerical Analysis

1.1.1 Overview and Definitions

Numerical analysis is the branch of mathematics that develops and uses algorithms to obtain numerical solutions for problems that often lack closed-form solutions or where exact answers are impractical. Fundamental characteristics include:

- **Algorithms and Iteration:** Problems are solved via step-by-step procedures that use many basic arithmetic operations.
- **Approximation:** Methods yield approximate results that are sufficiently close to the true value.
- **Computational Efficiency and Stability:** The choice of algorithm balances speed and error sensitivity.

A **numerical algorithm** is a finite procedure that takes input data (which may be inexact) and performs elementary arithmetic operations to generate an approximate solution. For example, the problem of approximating the square root of 2 can be solved using an iterative approach related to Newton's method.

1.1.2 Detailed Examples

Example 1.1: Approximating $\sqrt{2}$ with an Iterative Method A common iterative formula (a special case of Newton's method) for finding the square root of 2 is:

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right)$$

Choosing an initial guess $x_0 = 1.5$:

- **Iteration 1:**

$$x_1 = \frac{1}{2} \left(1.5 + \frac{2}{1.5} \right) \approx \frac{1}{2}(1.5 + 1.3333) \approx 1.4167$$

- **Iteration 2:**

$$x_2 = \frac{1}{2} \left(1.4167 + \frac{2}{1.4167} \right) \approx 1.4142$$

Thus, after only a few iterations, the method converges to approximately 1.4142, which is $\sqrt{2}$ to four decimal places.

Example 1.2: Approximating a Function Value Using a Taylor Series Consider using the Taylor series for the cosine function. For small x , we have:

$$\cos x \approx 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

For $x = 0.5$:

$$\cos 0.5 \approx 1 - \frac{0.25}{2} + \frac{0.0625}{24} \approx 1 - 0.125 + 0.00260 \approx 0.8776$$

This example shows how series approximations are used to estimate values when exact calculations are difficult.

1.2 Part 1.2: Error Analysis

1.2.1 Overview and Definitions

In numerical analysis, quantifying the error in an approximation is crucial. The two primary measures are:

- **Absolute Error:**

$$\text{Absolute Error} = |\text{True Value} - \text{Approximation}|$$

- **Relative Error:**

$$\text{Relative Error} = \frac{|\text{True Value} - \text{Approximation}|}{|\text{True Value}|}$$

Other important concepts include:

- **Truncation Error:** The error incurred by terminating an infinite process (such as a series) after a finite number of terms.
- **Roundoff Error:** The error due to finite precision in computer arithmetic.
- **Forward and Backward Error Analysis:** Analysis methods assessing, respectively, how far the computed result is from the true result and what small perturbations in the input could yield the computed result.

1.2.2 Detailed Examples

Example 2.1: Rounding Error in a Decimal Approximation Assume the number 0.1492 is given as an approximation accurate to four decimal places. The true value lies in the interval $[0.14915, 0.14925]$. Then:

- The maximum absolute error is approximately ± 0.00005 (half a unit in the fourth decimal place).
- The relative error is:

$$\frac{0.00005}{0.1492} \approx 0.000335 \quad (\text{or about } 0.0335\%)$$

Example 2.2: Error Propagation in Summation Suppose you add 100 numbers, each with a maximum error of ± 0.005 . In the worst case, the errors add up:

$$\text{Total Error} \leq 100 \times 0.005 = 0.5.$$

Even if each individual error is small, the cumulative error in a long summation may become significant.

Reputable sources: For further reading on error analysis, see Schaum's Outline of Numerical Analysis (Second Edition) and online resources such as the [Wikipedia article on Numerical Analysis](#).

2 Lecture 2: Newton-Raphson Method and Interpolation

2.1 Part 2.1: Newton-Raphson Method for Non-Linear Equations

2.1.1 Overview and Definitions

The **Newton-Raphson Method** is an iterative algorithm used to approximate the roots of a real-valued function $f(x)$. It is based on the first-order Taylor series expansion:

$$f(x) \approx f(x_n) + f'(x_n)(x - x_n).$$

Setting $f(x) = 0$ and solving for x , we obtain:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

The method converges *quadratically* under suitable conditions (for instance, when $f'(x)$ is non-zero near the root) and requires a good initial approximation.

2.1.2 Detailed Examples

Example 3.1: Solving $f(x) = x^2 - 2$ (Finding $\sqrt{2}$) Let $f(x) = x^2 - 2$ and $f'(x) = 2x$. Choose the initial guess $x_0 = 1.5$:

- **Iteration 1:**

$$x_1 = 1.5 - \frac{1.5^2 - 2}{2 \times 1.5} = 1.5 - \frac{0.25}{3} \approx 1.4167.$$

- **Iteration 2:**

$$x_2 = 1.4167 - \frac{(1.4167)^2 - 2}{2 \times 1.4167} \approx 1.4142.$$

The result converges quickly to $\sqrt{2}$.

Example 3.2: Solving $f(x) = \cos x - x$ Here, $f(x) = \cos x - x$ and $f'(x) = -\sin x - 1$. Choose an initial guess $x_0 = 0.5$:

- **Iteration 1:**

$$\begin{aligned} f(0.5) &= \cos 0.5 - 0.5 \approx 0.8776 - 0.5 = 0.3776, \\ f'(0.5) &\approx -\sin 0.5 - 1 \approx -0.4794 - 1 = -1.4794, \\ x_1 &= 0.5 - \frac{0.3776}{-1.4794} \approx 0.5 + 0.2553 = 0.7553. \end{aligned}$$

- **Iteration 2:** The next iteration further refines the solution.

This iterative process refines the approximation to the fixed point, where $\cos x = x$.

2.2 Part 2.2: Interpolation and Polynomial Approximation

Interpolation involves replacing a set of data points with a continuous polynomial function that passes through them exactly. Two common techniques are discussed below.

2.2.1 (a) Lagrange Interpolation Polynomial

Definition: Given $n + 1$ distinct data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, the **Lagrange interpolation polynomial** is defined as:

$$p(x) = \sum_{i=0}^n y_i L_i(x),$$

where

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}.$$

This polynomial is unique and exactly passes through the given points.

Example 4.1: Lagrange Interpolation with Three Points Given the data points: $(0, 1)$, $(1, 3)$, and $(2, 2)$, the basis polynomials are computed as:

$$L_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2},$$

$$L_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)} = -x(x-2),$$

$$L_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}.$$

The interpolating polynomial becomes:

$$p(x) = 1 \cdot \frac{(x-1)(x-2)}{2} + 3 \cdot [-x(x-2)] + 2 \cdot \frac{x(x-1)}{2}.$$

Simplify further to obtain the explicit form of $p(x)$.

Example 4.2: Error Estimation in Lagrange Interpolation The error in the Lagrange interpolation for a smooth function $y(x)$ is given by:

$$y(x) - p(x) = \frac{y^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i),$$

where ξ is some number in the interval containing the x_i . This formula provides an upper bound on the interpolation error.

2.2.2 (b) Newton Polynomial Interpolation

Definition: Newton's interpolation uses divided differences to express the polynomial in an incremental manner:

$$p(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n(x - x_0)(x - x_1) \cdots (x - x_{n-1}),$$

where the coefficients a_i are determined by the divided differences:

$$a_0 = y_0, \quad a_1 = f[x_0, x_1], \quad a_2 = f[x_0, x_1, x_2], \quad \dots$$

Example 4.3: Newton Interpolation with Three Points For the same data points $(0, 1)$, $(1, 3)$, $(2, 2)$:

- Compute $a_0 = y_0 = 1$.
- The first divided difference:

$$a_1 = f[x_0, x_1] = \frac{3-1}{1-0} = 2.$$

- Next, compute the second divided difference:

$$f[x_1, x_2] = \frac{2-3}{2-1} = -1, \quad a_2 = f[x_0, x_1, x_2] = \frac{-1-2}{2-0} = -1.5.$$

Thus, the Newton interpolation polynomial is:

$$p(x) = 1 + 2x - 1.5x(x-1).$$

Example 4.4: Evaluating the Newton Polynomial at $x = 1.5$ For $x = 1.5$:

- Calculate the factors:

$$(x - x_0) = 1.5 - 0 = 1.5, \quad (x - x_1) = 1.5 - 1 = 0.5.$$

- Substitute into the polynomial:

$$p(1.5) = 1 + 2(1.5) - 1.5(1.5)(0.5) = 1 + 3 - 1.125 = 2.875.$$

This gives the approximate interpolated value at $x = 1.5$.

Reputable sources: For more information on interpolation techniques, consult Schaum's Outline of Numerical Analysis (Second Edition) and online resources such as:

- [Newton-Raphson Method](#)
- [Lagrange Polynomial](#)
- [Divided Differences](#)

Final Notes

These two lectures have been structured to provide:

- A clear introduction to numerical analysis and error analysis, with detailed examples.
- An application of numerical methods to solve non-linear equations using the Newton-Raphson method.
- An explanation and construction of interpolating polynomials via both Lagrange and Newton methods.

For a thorough study, refer to Schaum's Outline of Numerical Analysis, Second Edition by Francis Scheid and the additional online resources noted above.

Citations: Schaum's Outline of Numerical Analysis, Second Edition, Francis Scheid, McGraw-Hill, 1989. Additional links are provided in the document via the hyperref package.

Q.1

Method 2

$$P(R \cup W) = P(B') = 1 - P(B) = 1 - \frac{1}{3} = \frac{2}{3} \text{ by part (c).}$$

Method 3

Since events R and W are mutually exclusive, it follows from (4), page 5, that

$$P(R \cup W) = P(R) + P(W) = \frac{2}{5} + \frac{4}{15} = \frac{2}{3}$$

A ball is drawn at random from a box containing 6 red balls, 4 white balls, and 5 blue balls. Determine the probability that it is (a) red, (b) white, (c) blue, (d) not red, (e) red or white.

(a) **Method 1**

Let R , W , and B denote the events of drawing a red ball, white ball, and blue ball, respectively. Then

$$P(R) = \frac{\text{ways of choosing a red ball}}{\text{total ways of choosing a ball}} = \frac{6}{6 + 4 + 5} = \frac{6}{15} = \frac{2}{5}$$

Method 2

Our sample space consists of $6 + 4 + 5 = 15$ sample points. Then if we assign equal probabilities $1/15$ to each sample point, we see that $P(R) = 6/15 = 2/5$, since there are 6 sample points corresponding to "red ball."

(b) $P(W) = \frac{4}{6 + 4 + 5} = \frac{4}{15}$

(c) $P(B) = \frac{5}{6 + 4 + 5} = \frac{5}{15} = \frac{1}{3}$

(d) $P(\text{not red}) = P(R') = 1 - P(R) = 1 - \frac{2}{5} = \frac{3}{5}$ by part (a).

(e) **Method 1**

$$\begin{aligned} P(\text{red or white}) &= P(R \cup W) = \frac{\text{ways of choosing a red or white ball}}{\text{total ways of choosing a ball}} \\ &= \frac{6 + 4}{6 + 4 + 5} = \frac{10}{15} = \frac{2}{3} \end{aligned}$$

This can also be worked using the sample space as in part (a).

Q.2

A fair die is tossed twice. Find the probability of getting a 4, 5, or 6 on the first toss and a 1, 2, 3, or 4 on the second toss.

Let A_1 be the event “4, 5, or 6 on first toss,” and A_2 be the event “1, 2, 3, or 4 on second toss.” Then we are looking for $P(A_1 \cap A_2)$.

Method 1

$$P(A_1 \cap A_2) = P(A_1)P(A_2|A_1) = P(A_1)P(A_2) = \left(\frac{3}{6}\right)\left(\frac{4}{6}\right) = \frac{1}{3}$$

We have used here the fact that the result of the second toss is *independent* of the first so that $P(A_2|A_1) = P(A_2)$. Also we have used $P(A_1) = 3/6$ (since 4, 5, or 6 are 3 out of 6 equally likely possibilities) and $P(A_2) = 4/6$ (since 1, 2, 3, or 4 are 4 out of 6 equally likely possibilities).

Method 2

Each of the 6 ways in which a die can fall on the first toss can be associated with each of the 6 ways in which it can fall on the second toss, a total of $6 \cdot 6 = 36$ ways, all equally likely.

Each of the 3 ways in which A_1 can occur can be associated with each of the 4 ways in which A_2 can occur to give $3 \cdot 4 = 12$ ways in which both A_1 and A_2 can occur. Then

$$P(A_1 \cap A_2) = \frac{12}{36} = \frac{1}{3}$$

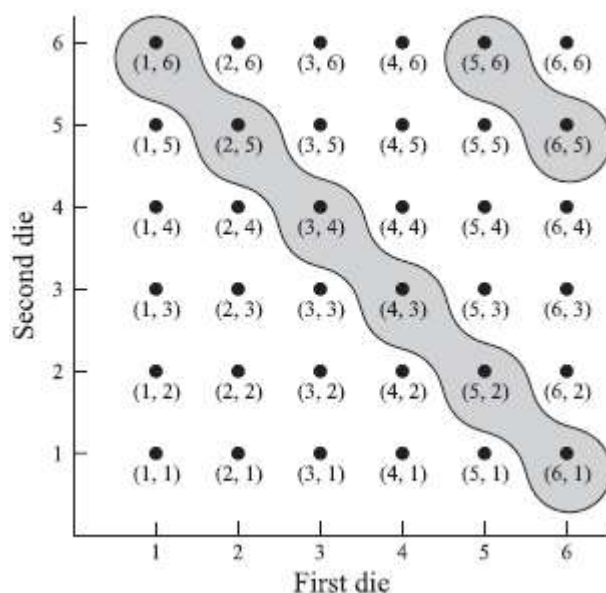
This shows directly that A_1 and A_2 are independent since

$$P(A_1 \cap A_2) = \frac{1}{3} = \left(\frac{3}{6}\right)\left(\frac{4}{6}\right) = P(A_1)P(A_2)$$

Q.3

Find the probability of not getting a 7 or 11 total on either of two tosses of a pair of fair dice.

The sample space for each toss of the dice is shown in Fig. 1-9. For example, (5, 2) means that 5 comes up on the first die and 2 on the second. Since the dice are fair and there are 36 sample points, we assign probability $1/36$ to each.



If we let A be the event “7 or 11,” then A is indicated by the circled portion in Fig. 1-9. Since 8 points are included, we have $P(A) = 8/36 = 2/9$. It follows that the probability of no 7 or 11 is given by

$$P(A') = 1 - P(A) = 1 - \frac{2}{9} = \frac{7}{9}$$

Using subscripts 1, 2 to denote 1st and 2nd tosses of the dice, we see that the probability of no 7 or 11 on either the first or second tosses is given by

$$P(A'_1)P(A'_2 | A'_1) = P(A'_1)P(A'_2) = \left(\frac{7}{9}\right)\left(\frac{7}{9}\right) = \frac{49}{81}$$

Q.4

Two cards are drawn from a well-shuffled ordinary deck of 52 cards. Find the probability that they are both aces if the first card is (a) replaced, (b) not replaced.

Method 1

Let A_1 = event “ace on first draw” and A_2 = event “ace on second draw.” Then we are looking for $P(A_1 \cap A_2) = P(A_1)P(A_2 | A_1)$.

- (a) Since for the first drawing there are 4 aces in 52 cards, $P(A_1) = 4/52$. Also, if the card is replaced for the second drawing, then $P(A_2 | A_1) = 4/52$, since there are also 4 aces out of 52 cards for the second drawing. Then

$$P(A_1 \cap A_2) = P(A_1)P(A_2 | A_1) = \left(\frac{4}{52}\right)\left(\frac{4}{52}\right) = \frac{1}{169}$$

- (b) As in part (a), $P(A_1) = 4/52$. However, if an ace occurs on the first drawing, there will be only 3 aces left in the remaining 51 cards, so that $P(A_2 | A_1) = 3/51$. Then

$$P(A_1 \cap A_2) = P(A_1)P(A_2 | A_1) = \left(\frac{4}{52}\right)\left(\frac{3}{51}\right) = \frac{1}{221}$$

Method 2

- (a) The first card can be drawn in any one of 52 ways, and since there is replacement, the second card can also be drawn in any one of 52 ways. Then both cards can be drawn in $(52)(52)$ ways, all equally likely.

In such a case there are 4 ways of choosing an ace on the first draw and 4 ways of choosing an ace on the second draw so that the number of ways of choosing aces on the first and second draws is $(4)(4)$. Then the required probability is

$$\frac{(4)(4)}{(52)(52)} = \frac{1}{169}$$

- (b) The first card can be drawn in any one of 52 ways, and since there is no replacement, the second card can be drawn in any one of 51 ways. Then both cards can be drawn in $(52)(51)$ ways, all equally likely.

In such a case there are 4 ways of choosing an ace on the first draw and 3 ways of choosing an ace on the second draw so that the number of ways of choosing aces on the first and second draws is $(4)(3)$. Then the required probability is

$$\frac{(4)(3)}{(52)(51)} = \frac{1}{221}$$

Q.5

Three balls are drawn successively from the box of Problem 1.7. Find the probability that they are drawn in the order red, white, and blue if each ball is (a) replaced, (b) not replaced.

Let R_1 = event “red on first draw,” W_2 = event “white on second draw,” B_3 = event “blue on third draw.” We require $P(R_1 \cap W_2 \cap B_3)$.

- (a) If each ball is replaced, then the events are independent and

$$\begin{aligned} P(R_1 \cap W_2 \cap B_3) &= P(R_1)P(W_2 | R_1)P(B_3 | R_1 \cap W_2) \\ &= P(R_1)P(W_2)P(B_3) \\ &= \left(\frac{6}{6+4+5}\right)\left(\frac{4}{6+4+5}\right)\left(\frac{5}{6+4+5}\right) = \frac{8}{225} \end{aligned}$$

- (b) If each ball is not replaced, then the events are dependent and

$$\begin{aligned} P(R_1 \cap W_2 \cap B_3) &= P(R_1)P(W_2 | R_1)P(B_3 | R_1 \cap W_2) \\ &= \left(\frac{6}{6+4+5}\right)\left(\frac{4}{5+4+5}\right)\left(\frac{5}{5+3+5}\right) = \frac{4}{91} \end{aligned}$$

Q.6

Find the probability of a 4 turning up at least once in two tosses of a fair die.

Let A_1 = event “4 on first toss” and A_2 = event “4 on second toss.” Then

$$\begin{aligned} A_1 \cup A_2 &= \text{event “4 on first toss or 4 on second toss or both”} \\ &= \text{event “at least one 4 turns up,”} \end{aligned}$$

and we require $P(A_1 \cup A_2)$.

Method 1

Events A_1 and A_2 are not mutually exclusive, but they are independent. Hence, by (10) and (21),

$$\begin{aligned} P(A_1 \cup A_2) &= P(A_1) + P(A_2) - P(A_1 \cap A_2) \\ &= P(A_1) + P(A_2) - P(A_1)P(A_2) \\ &= \frac{1}{6} + \frac{1}{6} - \left(\frac{1}{6}\right)\left(\frac{1}{6}\right) = \frac{11}{36} \end{aligned}$$

Method 2

Then

$$\begin{aligned}
 P(\text{at least one 4 comes up}) + P(\text{no 4 comes up}) &= 1 \\
 P(\text{at least one 4 comes up}) &= 1 - P(\text{no 4 comes up}) \\
 &= 1 - P(\text{no 4 on 1st toss and no 4 on 2nd toss}) \\
 &= 1 - P(A'_1 \cap A'_2) = 1 - P(A'_1)P(A'_2) \\
 &= 1 - \left(\frac{5}{6}\right)\left(\frac{5}{6}\right) = \frac{11}{36}
 \end{aligned}$$

Method 3

Total number of equally likely ways in which both dice can fall = $6 \cdot 6 = 36$.

Also

- Number of ways in which A_1 occurs but not $A_2 = 5$
- Number of ways in which A_2 occurs but not $A_1 = 5$
- Number of ways in which both A_1 and A_2 occur = 1

Then the number of ways in which at least one of the events A_1 or A_2 occurs = $5 + 5 + 1 = 11$. Therefore,
 $P(A_1 \cup A_2) = 11/36$.

Q.7

One bag contains 4 white balls and 2 black balls; another contains 3 white balls and 5 black balls. If one ball is drawn from each bag, find the probability that (a) both are white, (b) both are black, (c) one is white and one is black.

Let W_1 = event “white ball from first bag,” W_2 = event “white ball from second bag.”

- (a) $P(W_1 \cap W_2) = P(W_1)P(W_2 | W_1) = P(W_1)P(W_2) = \left(\frac{4}{4+2}\right)\left(\frac{3}{3+5}\right) = \frac{1}{4}$
- (b) $P(W'_1 \cap W'_2) = P(W'_1)P(W'_2 | W'_1) = P(W'_1)P(W'_2) = \left(\frac{2}{4+2}\right)\left(\frac{5}{3+5}\right) = \frac{5}{24}$
- (c) The required probability is
- $$1 - P(W_1 \cap W_2) - P(W'_1 \cap W'_2) = 1 - \frac{1}{4} - \frac{5}{24} = \frac{13}{24}$$

Q.8

Box I contains 3 red and 2 blue marbles while Box II contains 2 red and 8 blue marbles. A fair coin is tossed. If the coin turns up heads, a marble is chosen from Box I; if it turns up tails, a marble is chosen from Box II. Find the probability that a red marble is chosen.

Let R denote the event “a red marble is chosen” while I and II denote the events that Box I and Box II are chosen, respectively. Since a red marble can result by choosing either Box I or II, we can use the results of Problem 1.14 with $A = R$, $A_1 = I$, $A_2 = II$. Therefore, the probability of choosing a red marble is

$$P(R) = P(I)P(R | I) + P(II)P(R | II) = \left(\frac{1}{2}\right)\left(\frac{3}{3+2}\right) + \left(\frac{1}{2}\right)\left(\frac{2}{2+8}\right) = \frac{2}{5}$$



Q.9/H.W.

Suppose in Problem Q.8 that the one who tosses the coin does not reveal whether it has turned up heads or tails (so that the box from which a marble was chosen is not revealed) but does reveal that a red marble was chosen. What is the probability that Box I was chosen (i.e., the coin turned up heads)?

Q.10/H.W.

A committee of 3 members is to be formed consisting of one representative each from labor, management, and the public. If there are 3 possible representatives from labor, 2 from management, and 4 from the public, determine how many different committees can be formed using (a) the fundamental principle of counting and (b) a tree diagram.